

# DP Computer Science: A 4.3.2 Supervised Learning — Classification

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## 1. Core Concepts & Learning Objectives

### Learning Intentions

By the end of this lesson, students will be able to:

- **Differentiate** between regression and classification as two types of supervised learning problems
  - **Explain** the operating principles of the **K-Nearest Neighbours (K-NN)** algorithm
  - **Explain** the operating principles of a **Decision Tree** algorithm
  - **Analyse** a simple classification problem and determine which algorithm (K-NN or Decision Tree) might be more suitable
  - **Describe** real-world applications for both K-NN and Decision Trees
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### Key Vocabulary

| Term                        | Definition                                                                                                                   |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Classification              | A supervised learning technique used to predict discrete categorical outcomes (labels)                                       |
| Regression                  | A supervised learning technique used to predict continuous numerical values                                                  |
| K-Nearest Neighbours (K-NN) | A distance-based classification algorithm that classifies new points based on the majority class of their nearest neighbours |
| Decision Tree               | A rule-based classification algorithm that uses a tree-like structure of decisions to classify data                          |
| Feature                     | A measurable property or characteristic of the data being observed                                                           |

| Term          | Definition                                                                |
|---------------|---------------------------------------------------------------------------|
| Label         | The class or category that the model predicts                             |
| K             | The number of neighbours considered in the K-NN algorithm                 |
| Root node     | The topmost node in a decision tree                                       |
| Decision node | A node in a decision tree where a decision is made based on a feature     |
| Leaf node     | A terminal node in a decision tree that provides the final classification |

## Approaches to Learning (ATL) Elements

| Skill                                                   | Description                                                                                                   |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Thinking (Conceptual Understanding & Logical Reasoning) | Understanding the intuitive logic behind K-NN and Decision Trees; tracing the logical path of a decision tree |
| Communication                                           | Clearly explaining the step-by-step process of a classification algorithm to a peer                           |

## 2. Regression vs. Classification

### The Key Difference

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REGRESSION vs. CLASSIFICATION

### REGRESSION

Predicts CONTINUOUS numerical values

Answer: "How much?"

Question: "How many points will this student  
score on the exam?"

Output: "75%" (a number)

### CLASSIFICATION

Predicts DISCRETE CATEGORIES (labels)

Answer: "What category?"

Question: "Is this email spam or not spam?"

Output: "Spam" or "Not Spam" (a label)

## Quick Comparison

| Aspect    | Regression          | Classification                                  |
|-----------|---------------------|-------------------------------------------------|
| Output    | Continuous number   | Discrete category/label                         |
| Question  | "How much?"         | "What category?"                                |
| Example   | Predict house price | Predict if house is 'expensive' or 'affordable' |
| Algorithm | Linear Regression   | K-NN, Decision Tree                             |

## 3. K-Nearest Neighbours (K-NN)

# What is K-NN?

K-Nearest Neighbours (K-NN) is a distance-based classification algorithm. It classifies a new data point based on the majority class of its nearest neighbours.

## The "Guilt by Association" Analogy

*"You are who your friends are. A new point is classified by looking at its closest neighbours and taking a majority vote."*

## How K-NN Works

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HOW K-NN WORKS

Step 1: Choose a value for k (number of neighbours)

Step 2: Calculate the distance from the new point to all existing points

Step 3: Find the k nearest neighbours

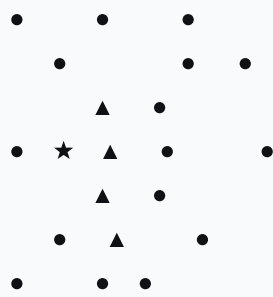
Step 4: Take a majority vote among the k neighbours

Step 5: Assign the majority class to the new point

## Visual Example

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K-NN VISUAL EXAMPLE



★ = New point to classify  
● = Existing data points

k = 3: The 3 nearest neighbours vote  
k = 5: The 5 nearest neighbours vote

● Blue neighbours: 2      ● Red neighbours: 1  
Classify as ● BLUE!

## The Importance of K

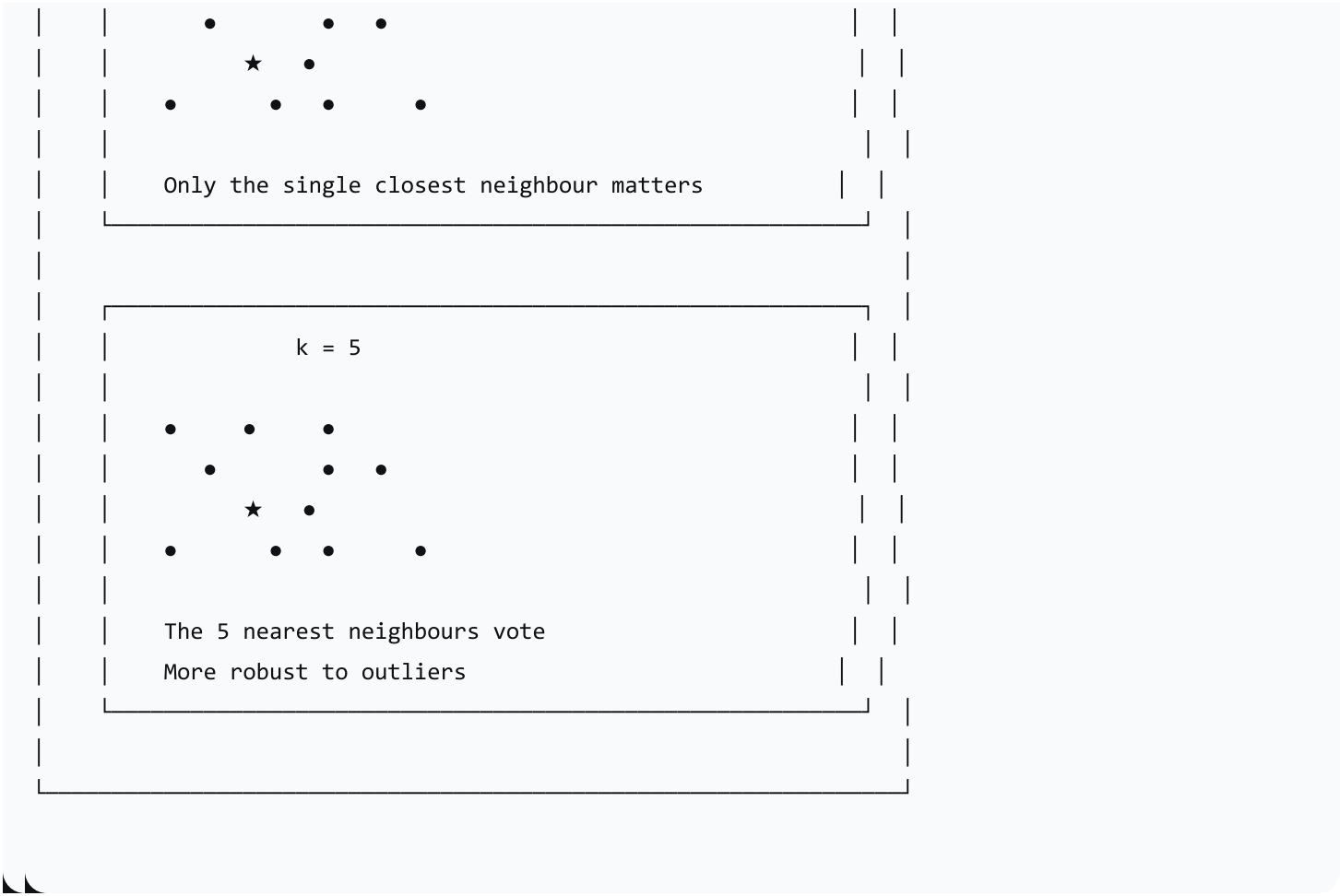
| Value of k             | Effect                                |
|------------------------|---------------------------------------|
| Small k (e.g., 1)      | Sensitive to noise; can overfit       |
| Medium k (e.g., 5, 10) | Balanced; good for most cases         |
| Large k (e.g., 100)    | Smooths over variations; may underfit |

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THE EFFECT OF DIFFERENT k VALUES

k = 1





## Real-World Applications of K-NN

| Application             | Description                                                     |
|-------------------------|-----------------------------------------------------------------|
| Collaborative Filtering | "Users who liked these movies also liked..."                    |
| Recommendation Systems  | Amazon: "Customers who bought this also bought..."              |
| Image Recognition       | Classify images by comparing to similar images                  |
| Medical Diagnosis       | Classify patients based on similarity to previous cases         |
| Fraud Detection         | Identify fraudulent transactions by comparing to known patterns |

## K-NN in Summary

| Aspect     | Description                                                                    |
|------------|--------------------------------------------------------------------------------|
| Type       | Supervised learning                                                            |
| Approach   | Distance-based                                                                 |
| Strengths  | Simple to understand; no training phase; works well with small datasets        |
| Weaknesses | Slow for large datasets; sensitive to irrelevant features; requires choosing k |

## 4. Decision Trees

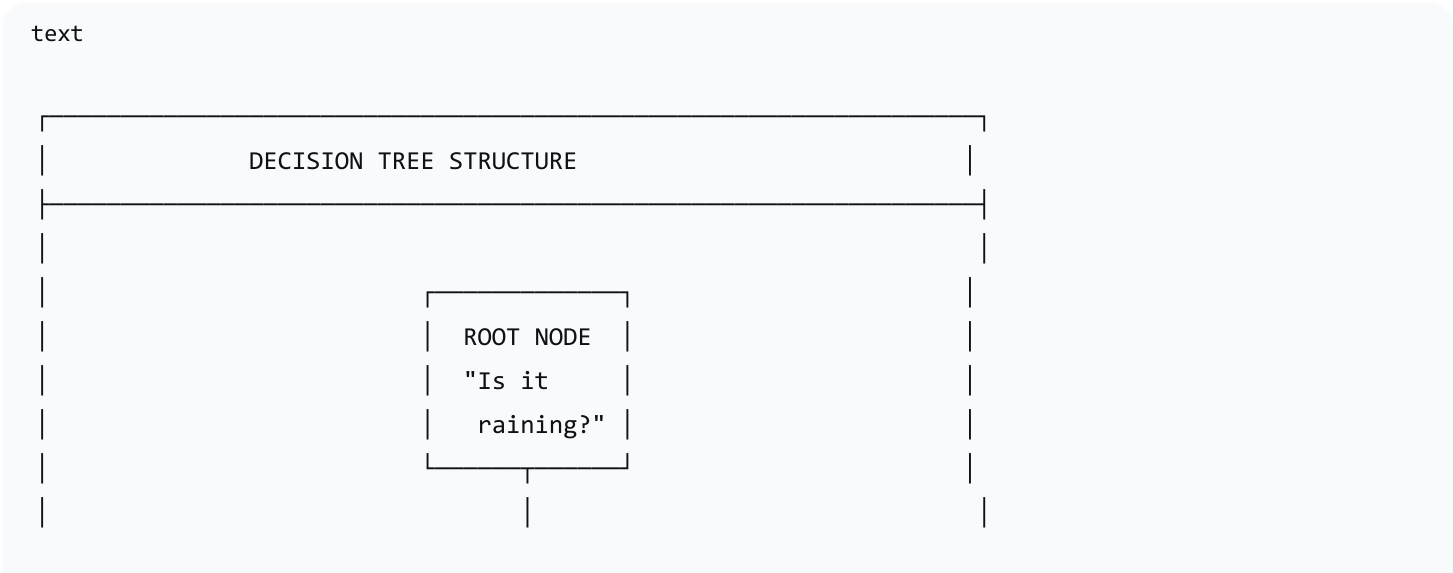
### What is a Decision Tree?

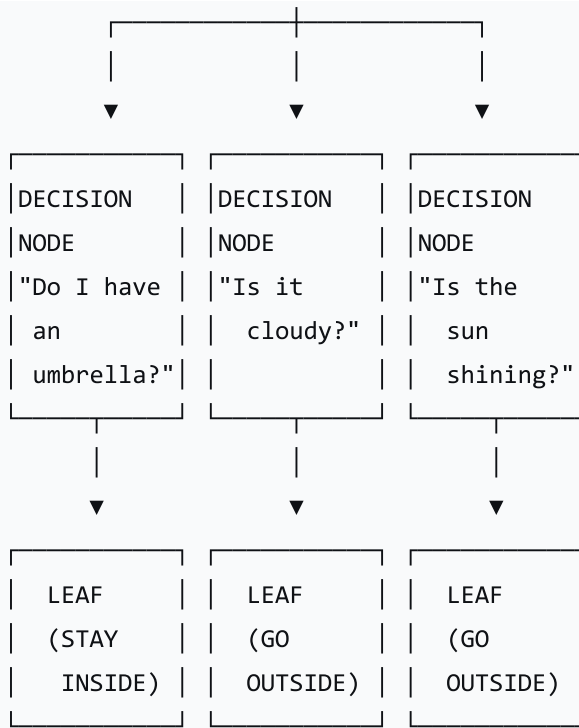
A Decision Tree is a rule-based classification algorithm that uses a tree-like structure of decisions to classify data. It mimics human decision-making by asking a series of questions.

### The Flowchart Analogy

"A decision tree is like a flowchart—you answer a series of questions, each leading to the next, until you reach a final classification."

### Decision Tree Structure





Root Node: The starting point

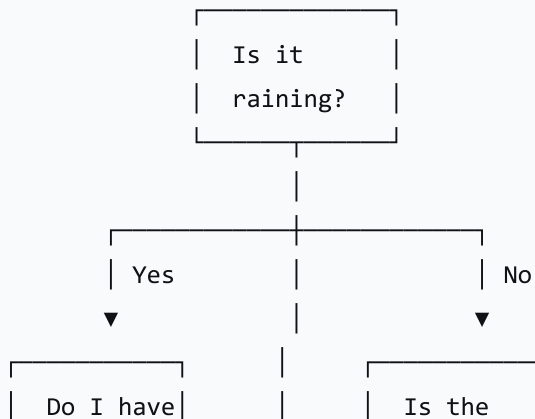
Decision Nodes: Points where a decision is made

Leaf Nodes: Final classification (no further decisions)

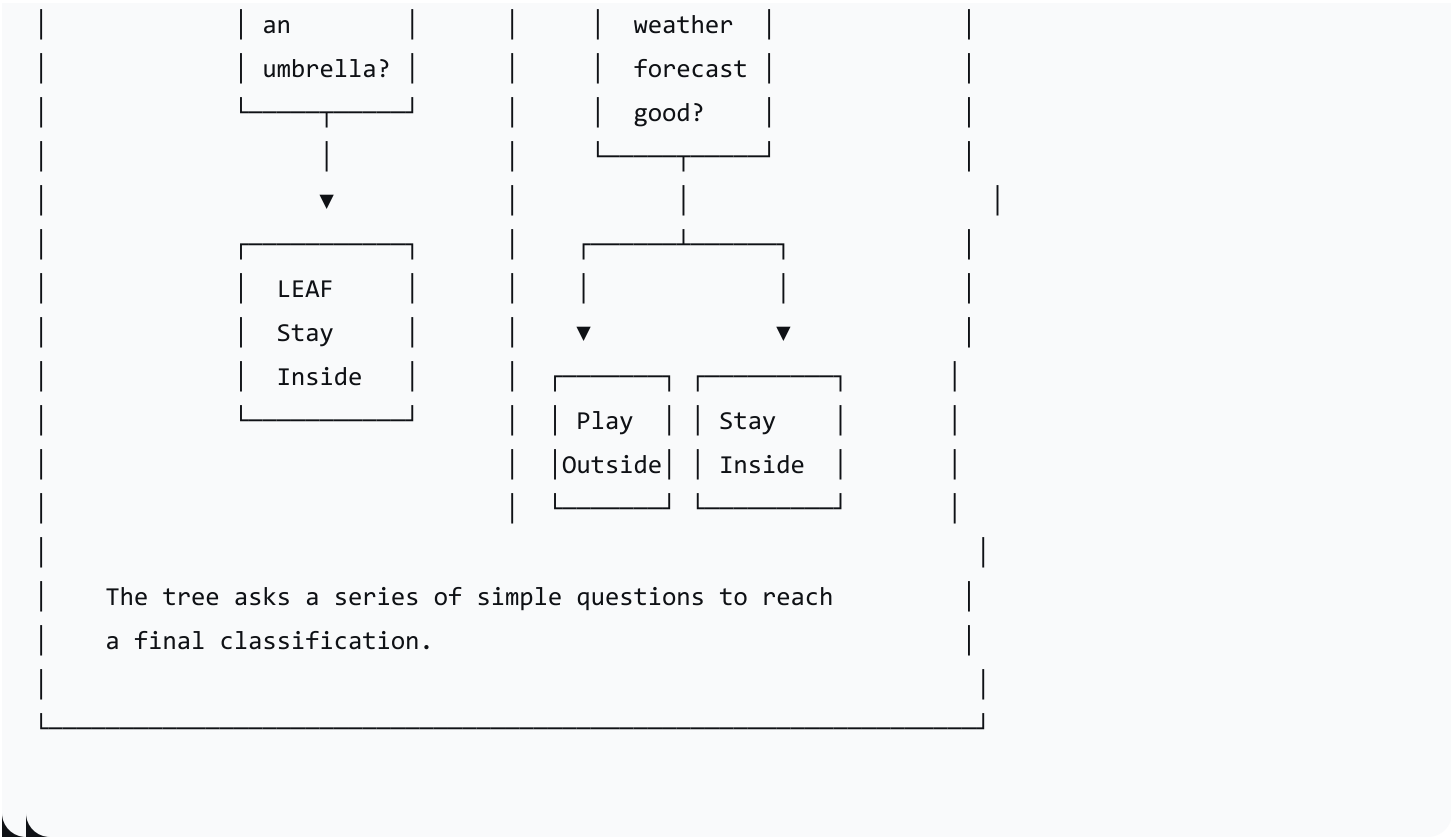
## Example: "Should I Go Outside to Play?"

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### DECISION TREE EXAMPLE







## Decision Tree Components

| Component     | Description                               | Example                       |
|---------------|-------------------------------------------|-------------------------------|
| Root Node     | The first decision point                  | "Is it raining?"              |
| Decision Node | An internal node where a decision is made | "Do I have an umbrella?"      |
| Leaf Node     | A terminal node; final classification     | "Stay Inside" or "Go Outside" |
| Branch        | The path from one node to the next        | Yes/No paths                  |

## Real-World Applications of Decision Trees

| Application       | Description                            |
|-------------------|----------------------------------------|
| Medical Diagnosis | Classify patients based on symptoms    |
| Credit Approval   | Determine if a loan should be approved |

| Application           | Description                                  |
|-----------------------|----------------------------------------------|
| Fraud Detection       | Identify potentially fraudulent transactions |
| Customer Segmentation | Group customers based on behaviour           |
| Troubleshooting       | Diagnose technical problems step by step     |

## Decision Trees in Summary

| Aspect     | Description                                                                 |
|------------|-----------------------------------------------------------------------------|
| Type       | Supervised learning                                                         |
| Approach   | Rule-based                                                                  |
| Strengths  | Easy to understand and explain; handles both numerical and categorical data |
| Weaknesses | Can overfit; sensitive to small changes in data                             |

## 5. K-NN vs. Decision Trees: Comparison

| Feature          | K-NN                           | Decision Tree                                       |
|------------------|--------------------------------|-----------------------------------------------------|
| Approach         | Distance-based                 | Rule-based                                          |
| Training Phase   | Lazy learner (no training)     | Eager learner (builds tree during training)         |
| Interpretability | Hard to explain decisions      | Very easy to explain decisions                      |
| Handles Data     | Works well with small datasets | Works well with both numerical and categorical data |
| Choosing k       | Must choose k value            | Tree depth is determined by algorithm               |

| Feature               | K-NN           | Decision Tree |
|-----------------------|----------------|---------------|
| Sensitive to Outliers | Yes            | Less so       |
| Learning              | Instance-based | Rule-based    |

## When to Use Which?

| Scenario                                        | Recommended Algorithm                       | Why?                                                      |
|-------------------------------------------------|---------------------------------------------|-----------------------------------------------------------|
| Small dataset; interpretability not important   | K-NN                                        | K-NN works well with small datasets                       |
| Medical diagnosis; interpretability is critical | Decision Tree                               | Easy to explain to doctors/patients                       |
| Data has mixed types (categorical + numerical)  | Decision Tree                               | Can handle mixed data types naturally                     |
| Data is high-dimensional                        | K-NN (or consider dimensionality reduction) | K-NN can work but may suffer from curse of dimensionality |
| Need a simple, understandable solution          | Decision Tree                               | Easy to visualise and explain                             |

## 6. Interactive Activity: "Human K-NN"

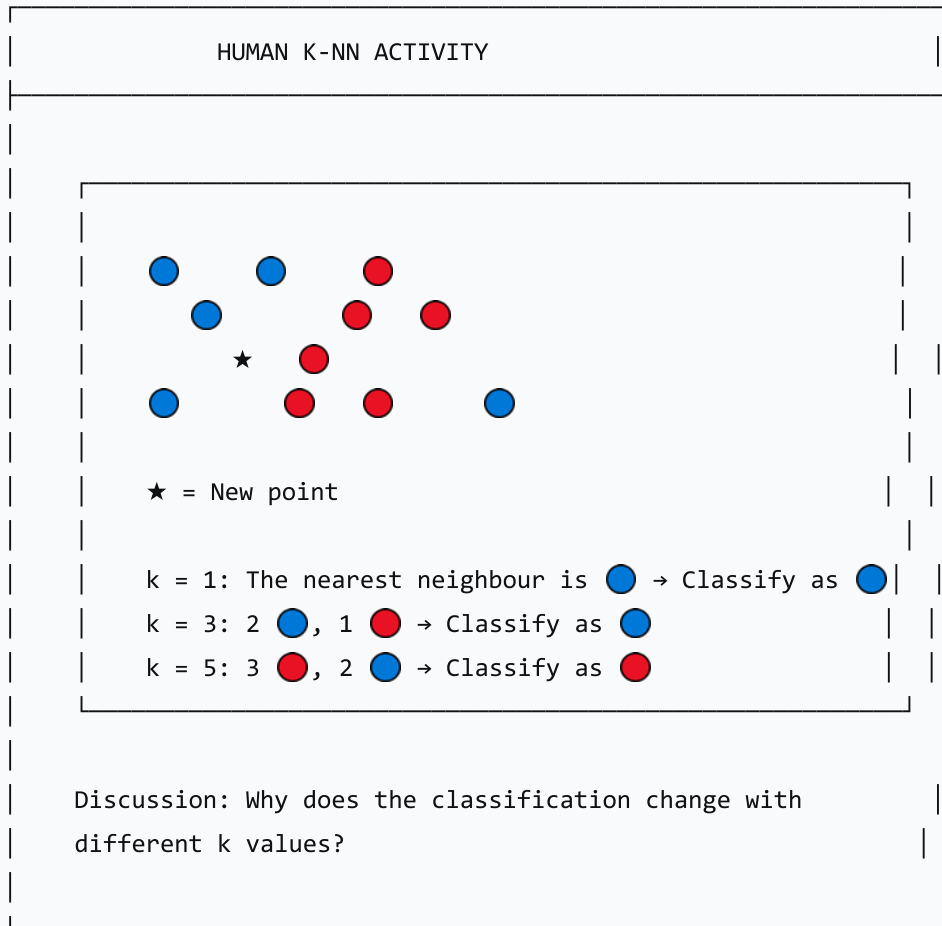
### Instructions

1. The teacher draws a **simple 2D plot** on the board
2. The plot contains **two classes** of points:
  -  Blue dots (Class A)
  -  Red dots (Class B)
3. A **new, unlabelled point** ★ is placed on the board
4. Students must classify the new point using K-NN:
  - If  $k = 1$ : What colour is the new point?

- If  $k = 3$ : What colour is the new point?
- If  $k = 5$ : What colour is the new point?

## Example Plot

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## Key Learning Points

| Observation | Insight                                           |
|-------------|---------------------------------------------------|
| $k = 1$     | Very sensitive to noise; may misclassify outliers |
| $k = 3$     | More stable; majority vote of 3 neighbours        |
| $k = 5$     | Even more stable; considers more neighbours       |

| Observation                              | Insight                      |
|------------------------------------------|------------------------------|
| Changing k changes the decision boundary | The choice of k is important |

## 7. Lesson Sequence

### Lesson Plan (60 minutes)

| Phase                   | Time   | Activity                                                                           |
|-------------------------|--------|------------------------------------------------------------------------------------|
| Recap & Introduction    | 10 min | Recap linear regression; introduce classification problem (flu vs. cold)           |
| Lecture: K-NN           | 15 min | "Guilt by Association" analogy; step-by-step process; real-world applications      |
| Interactive Activity    | 15 min | "Human K-NN" — classify points on the board with different k values                |
| Lecture: Decision Trees | 15 min | Flowchart analogy; structure (root, decision, leaf nodes); real-world applications |
| Consolidation           | 5 min  | Compare K-NN and Decision Trees; preview next lesson                               |

### Discussion Questions

| Question                                                        | Purpose                   |
|-----------------------------------------------------------------|---------------------------|
| "What is the difference between regression and classification?" | Differentiate concepts    |
| "How does K-NN classify a new point?"                           | Explain algorithm process |

| Question                                             | Purpose                    |
|------------------------------------------------------|----------------------------|
| "What happens if we change the value of k?"          | Understand importance of k |
| "What is the role of leaf nodes in a decision tree?" | Understand tree structure  |
| "When would you choose a decision tree over K-NN?"   | Compare algorithms         |

## 8. Assessment Activities

| Assessment Type      | Description                                                  |
|----------------------|--------------------------------------------------------------|
| Class Participation  | Observe engagement and contributions                         |
| Interactive Activity | Assess understanding of K-NN                                 |
| Exit Ticket          | Gauge understanding and identify areas needing clarification |

### Sample Exit Ticket Questions:

1. What is the difference between regression and classification?
2. How does the K-NN algorithm classify a new point?
3. What is the purpose of a leaf node in a decision tree?
4. Give one real-world application of a decision tree.

## 9. Differentiation

| Level     | Strategy                                                                                               |
|-----------|--------------------------------------------------------------------------------------------------------|
| Support   | Use clear analogies and visual examples; step through the "Human K-NN" activity carefully              |
| Challenge | Research how decision trees are built (splitting criteria); explore different distance metrics in K-NN |

# 10. Resources

- **Presentation** (Slide Deck)
- **eBook** (A4.3.2 with examples)
- **Worksheet:** Classification techniques (K-NN and Decision Trees)
- **Worksheet Answers**
- **Quizlet** (A4.3.2 vocabulary flashcards)

# 11. Summary: Key Takeaways

| Concept        | Key Point                                                         |
|----------------|-------------------------------------------------------------------|
| Regression     | Predicts continuous numbers (How much?)                           |
| Classification | Predicts discrete categories (What category?)                     |
| K-NN           | Distance-based; classifies by majority vote of nearest neighbours |
| K              | Number of neighbours; affects classification                      |
| Decision Tree  | Rule-based; uses tree of decisions to classify                    |
| Root Node      | First decision point                                              |
| Leaf Node      | Final classification                                              |

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| KEY REMINDERS |                                                       |
|---------------|-------------------------------------------------------|
| ✓             | Regression = predict a number                         |
| ✓             | Classification = predict a category                   |
| ✓             | K-NN = distance-based; "you are who your friends are" |
| ✓             | Decision Tree = rule-based; "flowchart of decisions"  |
| ✓             | K = number of neighbours in K-NN                      |
| ✓             | Root node = first decision in a decision tree         |

✓ Leaf node = final classification

*"Classification helps us answer 'what category?'—from diagnosing diseases to identifying spam, it's a powerful tool for making decisions."*